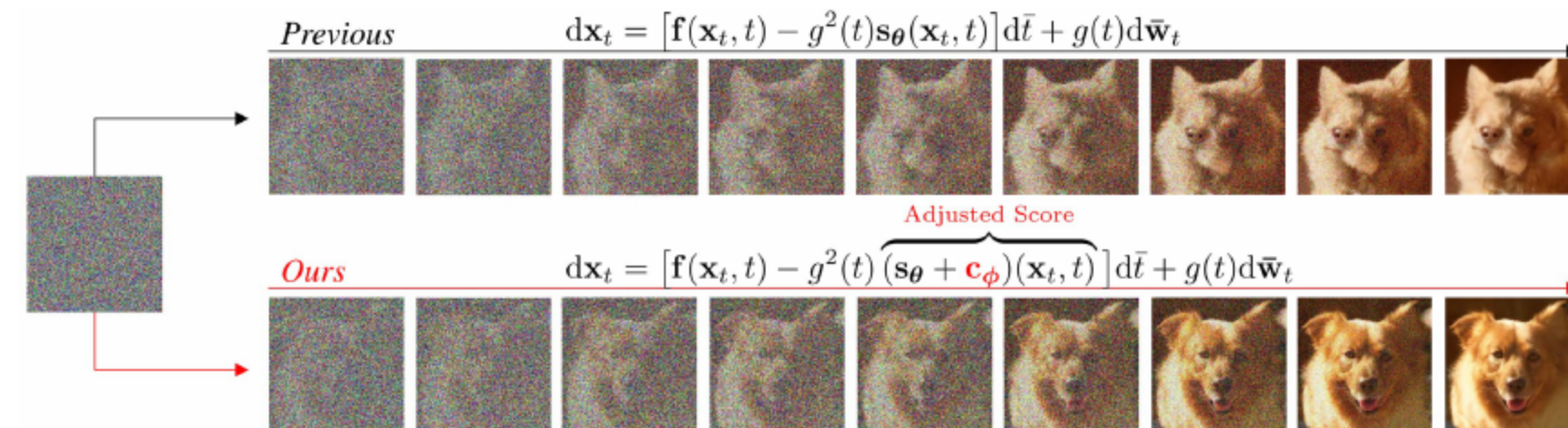


Refining Generative Process with Discriminator Guidance in Score-based Diffusion Models

Dongjun Kim, Yeongmin Kim, Se Jung Kwon,
Wanmo Kang, Il-Chul Moon

01 Why Improve Pre-trained Diffusion Models?

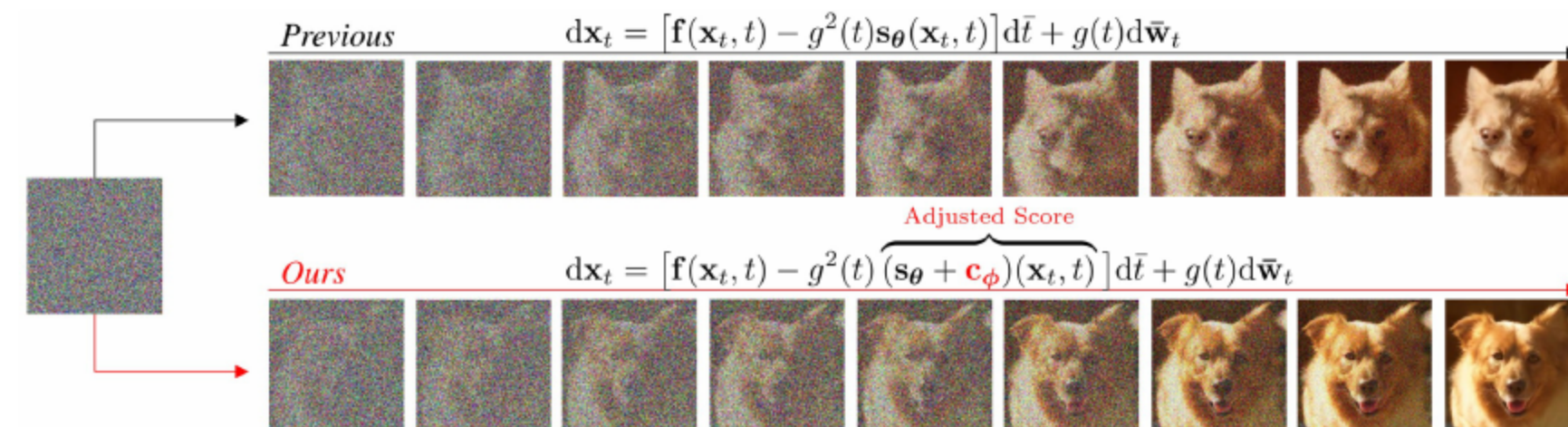


- Pre-trained diffusion scores deviate from data, reducing sample realism and diversity
- Full retraining is costly; finetuning risks overfitting and memorization
- Goal: lightweight, stable, sampling-time refinement without altering the score

01 Problem Setup And Objectives

- Frozen score $s_{\theta_{\infty}}$ may misalign with true data reverse process
- Objective: estimate a correction aligning model and data scores
- Apply correction at sampling to improve realism and diversity
- Avoid retraining and adversarial instability

02 Forward And Reverse SDEs

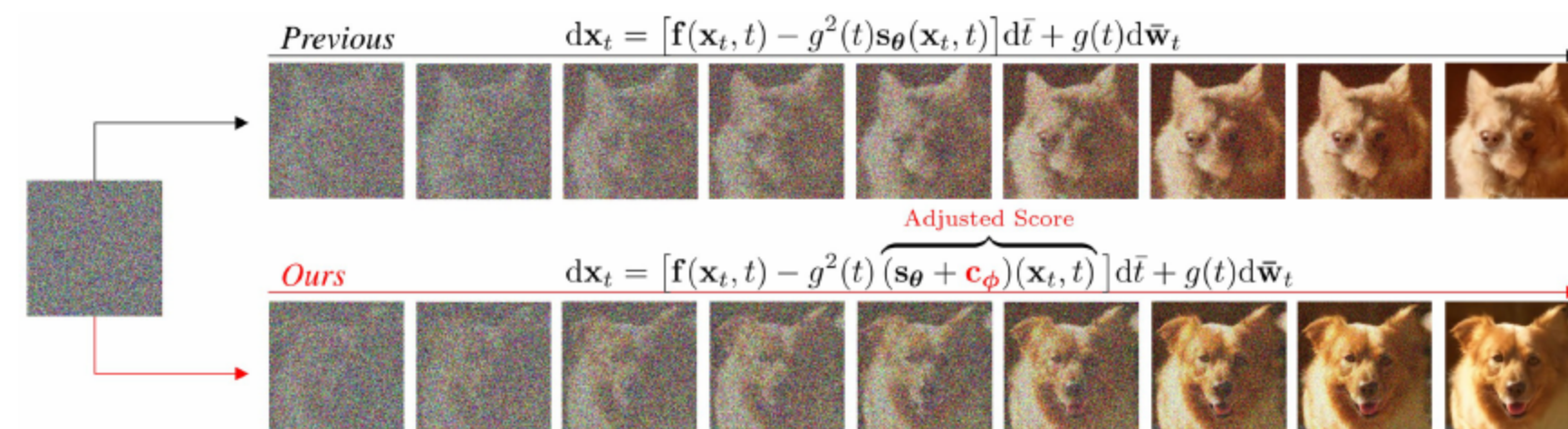


- Forward SDE perturbs data; reverse SDE uses data score for generation
- Train via denoising score matching to approximate $\nabla \log p_{\text{tr}}(x_t)$
- Score errors accumulate along trajectories, harming quality

02 Score Training Objectives

- Loss integrates time-weighted denoising errors; ties to KL minimization
- Alternate views: noise prediction and reconstruction losses
- Large-time score estimation remains hard; motivates sampling-time fixes

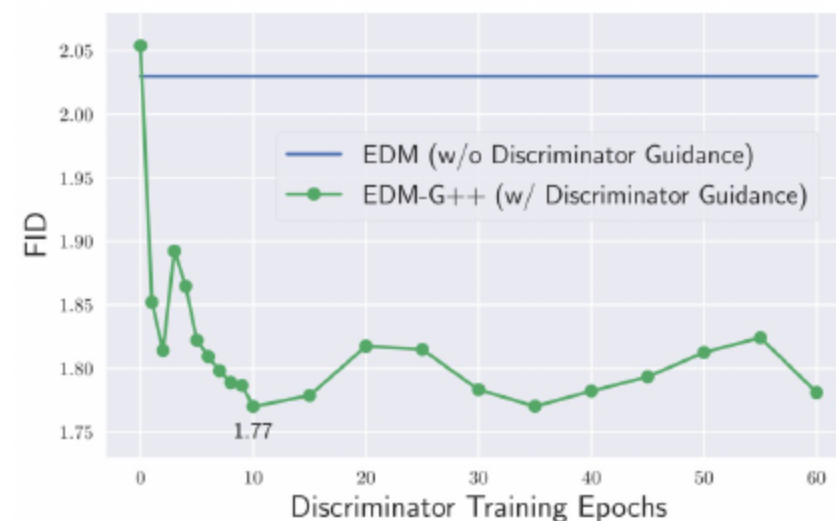
03 Discriminator Guidance For Score Correction



- Introduce Discriminator Guidance (DG) to correct frozen scores during sampling
- Train discriminator post score convergence; provides correction term
- Stable, fast training versus GANs; no generator updates

03 Theory And Practice

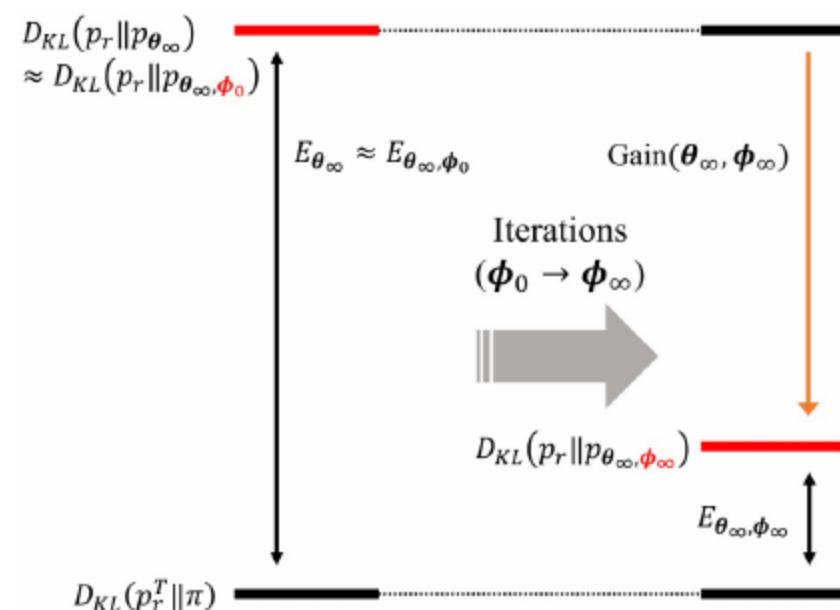
- Optimal discriminator recovers likelihood-ratio correction
- Adjusted score matches data score, reducing KL divergence
- DG achieves SOTA FID/recall with minimal overhead; complements Classifier Guidance



04 Correction Of Pre-trained Model Score

- Ideal correction $c(x_t, t) = \nabla \log[p_{tr}(x_t)/p_{t^{\infty}}(x_t)]$
- Adding c to $s_{\theta^{\infty}}$ aligns reverse SDE with true data process
- Reframes sampling as adjusting a fixed vector field

- Train time-conditioned discriminator over noise scales with BCE
- Recover likelihood ratio from optimal discriminator
- Use neural estimate $c\phi$ with schedule $w_DG(t)$ during sampling



PART 05

Technical Details And Algorithms

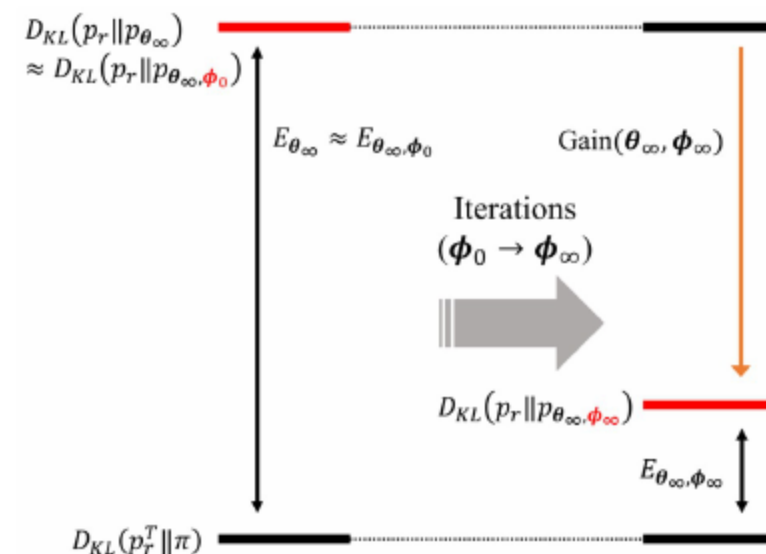
- Generate fake set via frozen score sampling
- Diffuse real and fake to time t ; optimize time-weighted BCE
- Iterate mini-batches until convergence; training is stable and quick

Sampler	γ_t	η	w_t^{CG}	w_t^{DG}
DDPM	0	1	0	0
DDIM	0	0	0	0
EDM	≥ 0	0	0	0
ADM-G	0	1	> 0	0
EDM-G++	≥ 0	0	0	> 0
ADM-G++	0	1	> 0	> 0

- Update reverse-step drift with $s\theta_\infty + w_DG(t)c\phi$
- Optionally add Classifier Guidance within unified weighted framework
- Applicable to DDPM, DDIM, and EDM samplers

05 Theoretical Guarantees And Gain

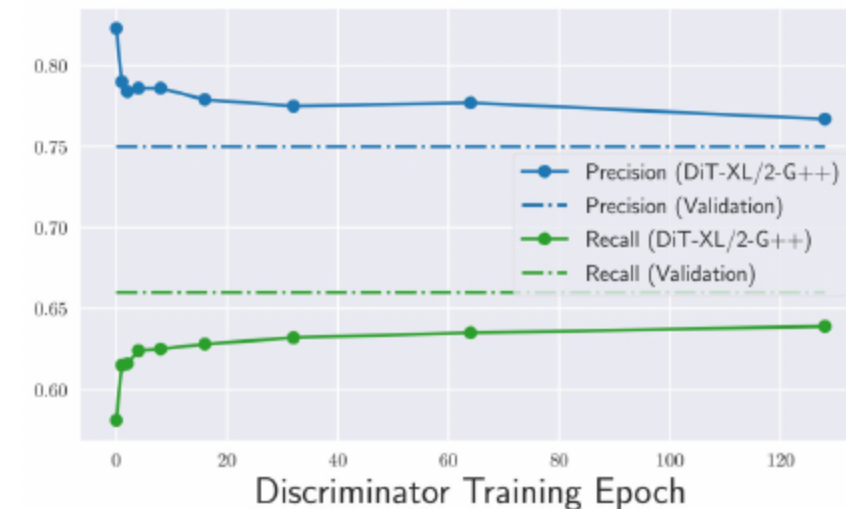
Discriminator	$E_{\theta_\infty, \phi}$	Gain
Blind $d_{\phi_b} (\equiv 0.5)$	E_{θ_∞}	0
Optimal d_{ϕ_*}	0	E_{θ_∞} (Maximum)
Untrained $d_{\phi_0} (\approx 0.5)$	$\approx E_{\theta_\infty}$	≈ 0
Trained d_{ϕ_∞}	$\ll E_{\theta_\infty}$	$\nearrow E_{\theta_\infty}$



- DG reduces KL by decreasing discriminator-adjusted score error
- As discriminator learns, gain monotonically increases
- At optimality, adjusted process recovers true reverse dynamics

05 Optimality And Diversity Control

- DG weight w controls fidelity-diversity trade-off
- Optimal discriminator enables tunable mixture-like behavior
- Empirically, schedules find sweet spots improving FID and recall



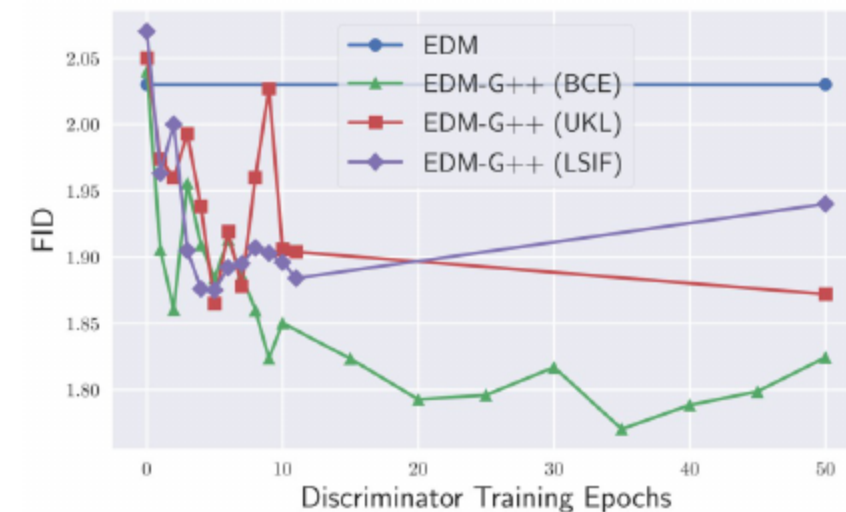
06 Relation To Classifier Guidance

- Classifier Guidance boosts class-consistency but may reduce recall
- DG complements by supervising realism
- Combining CG+DG mitigates precision-recall trade-offs



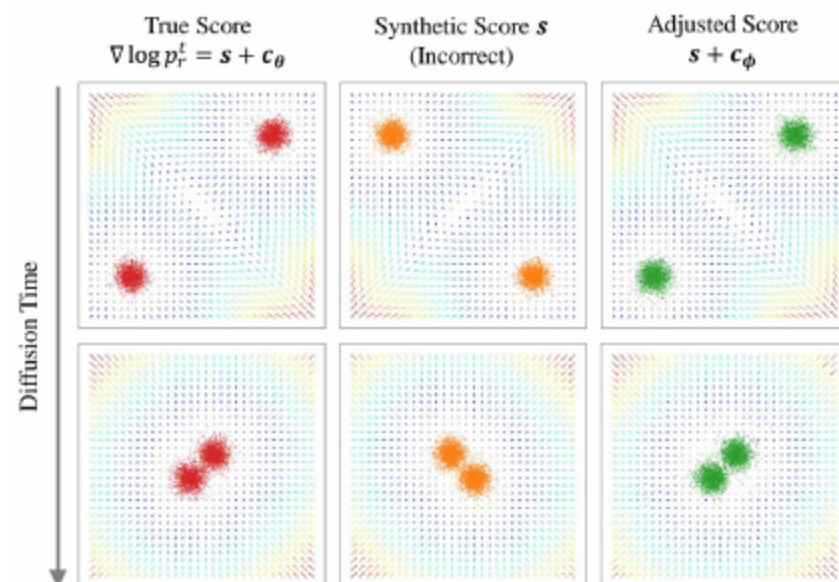
06 Links To NCE And Divergence Families

- DG aligns with time-varying NCE estimating likelihood ratios
- BCE relates to h-Bregman and f-divergences
- Alternative divergences can target perceptually plausible regions



07 Toy 2D Validation

- Learned correction aligns synthetic score with true data score
- Fixes trajectory misassignments; matches target densities visually



07 Image Generation Benchmarks

Model	Diffusion Space	NFE↓	Unconditional NLL↓	FID↓	Conditional FID↓
VDM (Kingma et al., 2021)	Data	1000	2.49	7.41	-
DDPM (Ho et al., 2020)	Data	1000	3.75	3.17	-
iDDPM (Nichol & Dhariwal, 2021)	Data	1000	3.37	2.90	-
Soft Truncation (Kim et al., 2022b)	Data	2000	2.91	2.47	-
INDM (Kim et al., 2022a)	Latent	2000	3.09	2.28	-
CLD-SGM (Dockhorn et al., 2022)	Data	312	3.31	2.25	-
NCSN++ (Song et al., 2020b)	Data	2000	3.45	2.20	-
LSGM (Vahdat et al., 2021)	Latent	138	3.43	2.10	-
NCSN++-G (Chao et al., 2022)	Data	2000	-	-	2.25
EDM (random seed)	Data	39	2.60	2.03	1.82
EDM (reported, manual seed)	Data	35	2.60	1.97	1.79
LSGM-G++	Latent	138	3.42	1.94	-
EDM-G++	Data	35	2.55	1.77	1.64

- Evaluate on CIFAR-10, CelebA/FFHQ 64×64, ImageNet 256×256
- Apply DG to EDM, LSGM, ADM, DiT with time-conditioned discriminators
- Report FID, sFID, IS, precision, recall, F1

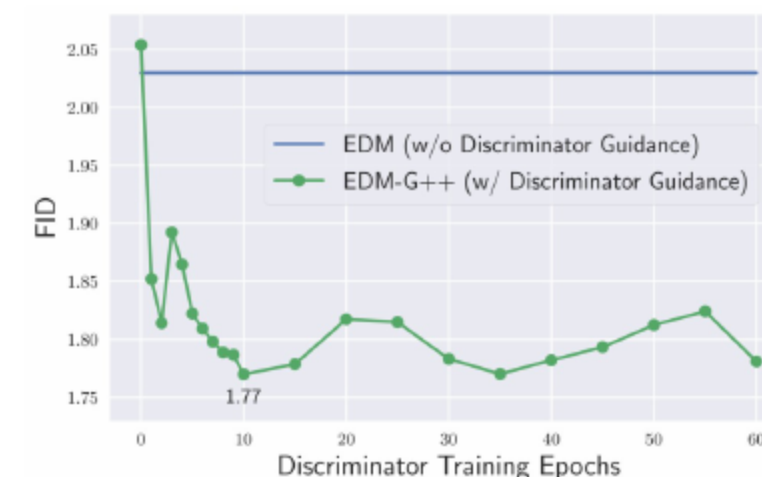
07 Compute And Efficiency

Model	Diffusion Space	FID↓	sFID↓	IS↑	Prec↑	Rec↑	F1↑
Validation Data		1.68	3.67	232.21	0.75	0.66	0.70
ADM (Dhariwal & Nichol, 2021)	Data	10.94	6.02	100.98	0.69	0.63	0.66
DiT-XL/2 (Peebles & Xie, 2022)	Latent	9.62	6.85	121.50	0.67	0.67	0.67
ADM-G (Dhariwal & Nichol, 2021)	Data	4.59	5.25	186.70	0.82	0.52	0.64
RIN (Jabri et al., 2022)	Data	4.51	-	161.00	-	-	-
LDM-4-G (Rombach et al., 2022)	Latent	3.60	-	247.67	0.87	0.48	0.62
RIN + schedule (Chen, 2023)	Data	3.52	-	186.20	-	-	-
simple diffusion (Hoogeboom et al., 2023)	Data	2.77	-	211.80	-	-	-
DiT-XL/2-G (Peebles & Xie, 2022)	Latent	2.27	4.60	278.24	0.83	0.57	0.68
ADM-G++	Data	3.18	4.53	255.74	0.84	0.53	0.66
DiT-XL/2-G++	Latent	1.83	5.16	281.53	0.78	0.64	0.70

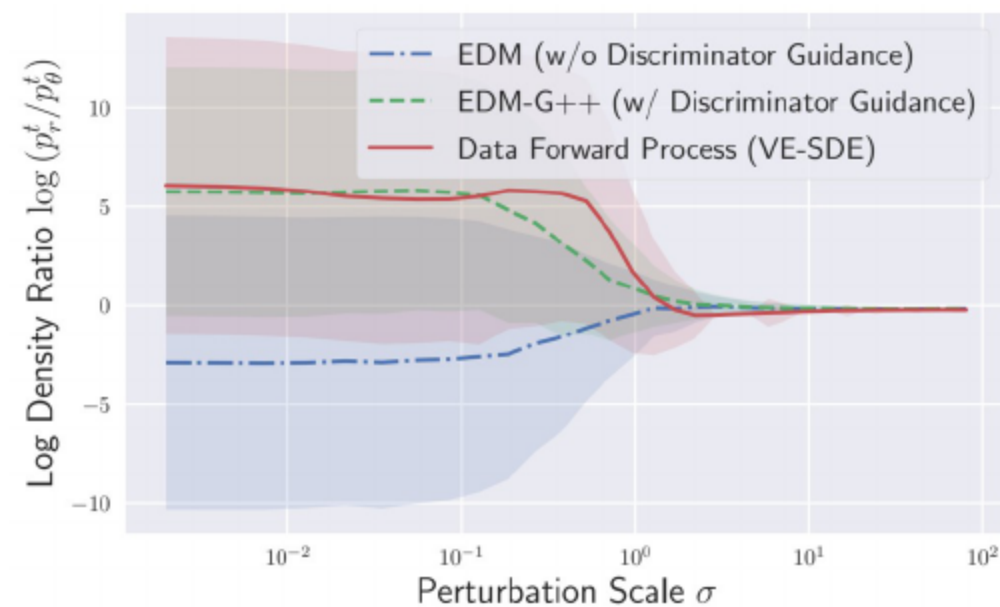
- Overhead: fake set generation + small discriminator training
- Converges quickly versus costly score training
- Modest extra NFEs during sampling

08 State-of-the-Art Performance

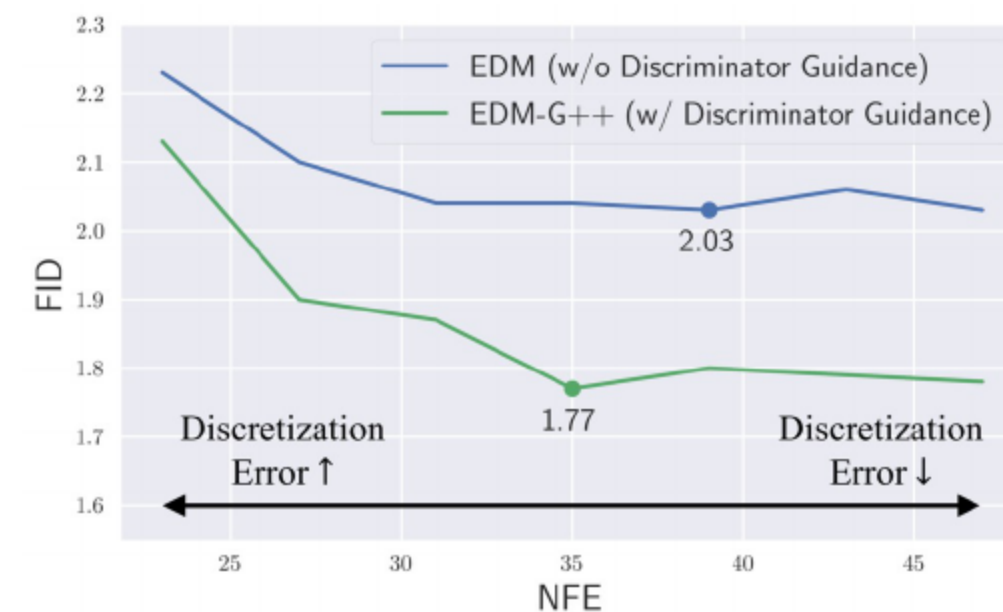
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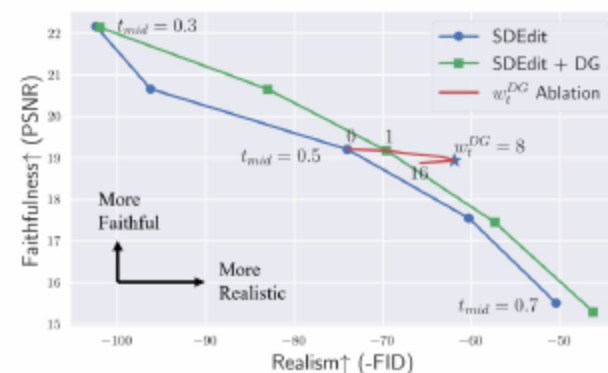
- DG sets SOTA on CIFAR-10 and face datasets
- ImageNet 256×256: FID 1.83, recall 0.64 with DiT-XL/2-G++
- Gains stem purely from guidance; base hyperparameters unchanged



- DG balances fidelity and coverage, mitigating overprecision
- Adjusted trajectories follow likelihood-ratio profiles of real data
- Diagnostics indicate improved score accuracy



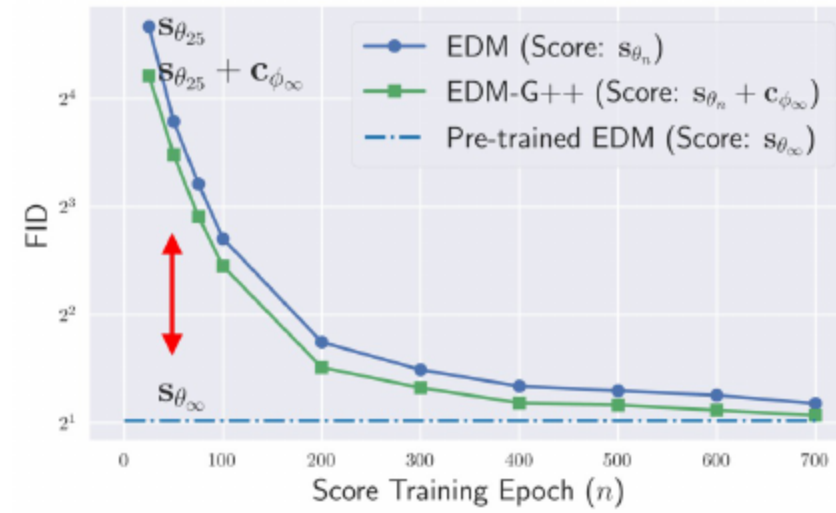
- Benefits shrink under very low NFE due to discretization
- DG weight schedules have sweet spots for FID and recall
- Alternative divergences yield comparable or better FIDs



- Build fake sets from translations; train discriminator vs. target domain
- DG improves realism while preserving faithfulness
- Enhances SDEdit baselines across noise schedules

- DG corrects frozen diffusion scores via likelihood-ratio estimation
- Improves realism/diversity with minimal cost and stable training
- Delivers SOTA and synergizes with Classifier Guidance

- Design principled divergences and joint min-min training
- Robust low-NFE regimes and solver integration
- Risks: stronger deepfakes; require safeguards and responsible use



THANKS!